Laboratoire d'Astrophysique de Marseille

Exploring the validity of the $log(R'_{hk})$ as an activity indicator in main-sequence stars

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Abstract

This study focuses mainly on the $log(R'_{HK})$, derived from the Ca II H & K emission lines, and attempts to use the Ca II IRT as an equivalent diagnostics tool for stellar activity.

The data used in this study is provided by VLT - UVES for the spectroscopy, and Kepler/K2 for the photometry. The set of spectroscopic data was specifically taken to be simultaneous to the photometry data, enabling better than ever comparison between the two.

During this study we find that the Ca II IRT is not particularly well correlated to the Ca II H & K without the use of a model. Moreover, we find that the H & K lines seem to be bad indicator of stellar activity for $log(R'_{HK}) < -4.6$.

I. Introduction

Nowadays, the hunt for exoplanets is more than ever increasing. Discoveries of planets outside the solar system are more and more numerous, and the need for better precision is at an all time high. This higher precision, made possible by state of the art technologies, is now mainly limited by the little knowledge we have on stellar activity.

Active stars may be interfering with measurements of high precision in the RV or transit methods, as the presence of spots or plages can induce false readings of radial velocity, or significantly alter transit luminosity (Dumusque et al. 2014). For those reasons, and until there are better capabilities to predict stellar activity, it has been customary to identify stars as active or inactive, using only the inactive stars for exoplanetary surveys.

This classification is done using the $log(R'_{HK})$, a processed value of the chromospheric emission situated in the Ca II H & K lines. This value has been widely accepted as being a good proxy for stellar activity. We will follow procedures similar to the ones used in Noyes et al. (1984), Lovis et al. (2011), Gomes da Silva et al. (2012) to process the Ca II H & K data into the $log(R'_{HK})$. The details of this will be explained in part III. a&b. We will also try a different approach to the commonplace one when it comes to the Ca II IRT, where we will not be using a model based method.

First we will describe briefly the data sets used in this study, then we will explain the processing done, presents the results, and finally discuss them.

II. Data sets

We use a sample of 81 main-sequence stars ranging from F5 to M3 for which we get data coming from the VLT-UVES spectrograph. This allows us to obtain not only the region of the Ca II H & K, but also the Ca II IRT, as UVES is the only spectrograph capable of capturing both at a resolution high enough (40000) to derive the $log(R')$ values of each regions.

These spectra were taken during the Kepler/K2 photometry campaign, allowing us to compare directly the photometric variability to the calculated $log(R'_{HK})$. The photometric data obtained through Kepler is especially interesting, as it isn't limited by atmospheric interferences, allowing unprecedented precision in very low variability stars.

After processing the spectra from UVES, we identified 15 binaries, which we will not be using in this study. Three stars were also not surveyed by K2, giving us a final sample of 63 stars to work with.

III. Data reduction

III. a. S index and calibration to Mount Wilson

To be able to derive the $log(R'_{HK})$, we first need to derive the historical S index value (Wilson 1978, Vaughan et al. 1980), which corresponds to the ratio of the flux inside the H & K lines to the continuum of the stars:

$$S = \alpha \cdot \frac{F_H + F_K}{F_B + F_V}$$

Where F_H and F_K are the total fluxes measured in the H & K lines, centered on 3933.664 Å and 3968.470 Å respectively, both with an FWHM of 1.09 Å. F_B and F_V are the fluxes used to define the continuum of the star, centered respectively around 3901.070 Å and 4001.070 Å, with a width of 20 Å. The α is a calibration constant, this value is either fixed at 2.30 or 2.40 (Duncan et al. 1991), or adjusted after calibration (Baliunas et al. 1995).

As for Lovis et al. (2011), we note that because of the way the MW telescope was designed, it let 8 times more flux in the continuum bandpasses. To account for this, we have to multiply the value of α, F_H and F_K by 8. Moreover, we also work using the mean fluxes per wavelength, that is $F_x^m = F_x/\Delta\lambda_x$, giving us the equation:

$$S = 8 \cdot \alpha \cdot \frac{1.09}{20.0} \cdot \frac{F_H^m + F_K^m}{F_B^m + F_V^m}$$

If considering α to be 2.3, we find a factor $\simeq 1$, giving us the final expression:

$$S \simeq \frac{F_H^m + F_K^m}{F_B^m + F_V^m}$$

We also measure and propagate the error on S due to noise on the detector, we follow a procedure similar to the one used in Robertson et al. (2013).

The value of S obtained this way now needs to be calibrated to the scale of MW, as both telescopes don't have the same calibrations, scattered light, or value of α (that is only assumed). To do so we find a sample of 12 stars that have been observed by both the MW and UVES, and compute the following table:

Star	S_{MW}	S_{UVES}
HD22049	0.510	0.043
HD156026	0.835	0.077
HD16160	0.230	0.021
HD17925	0.700	0.063
HD160346	0.305	0.025
HD114710	0.200	0.018
HD76151	0.260	0.021
HD190007	0.780	0.065
HD20630	0.345	0.031
HD149661	0.360	0.035
HD23249	0.137	0.012
HD155885	0.375	0.037

We use the values of S found in Noyes et al. (1984) for the S at MW. We then find a linear fit between our data and the one from MW:

$$S_{MW} = 11.1 \cdot S_{UVES} + 6.73 \cdot 10^{-3}$$

Once this scale is applied, we note that what we find seems to be in agreement with what was previously found on the values of S (**Fig. 1**).

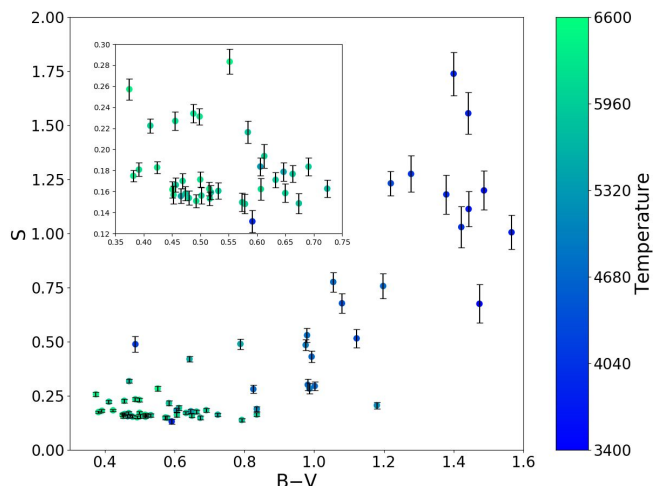


Fig. 1: *Measured S(H,K) scaled to the Mount-Wilson values as a function of B-V. The upper left part correspond to a zoom on the packed lowest values.*

We observe that S is more important for colder stars, which is coherent, as colder stars have less emission towards the blue side of the spectrum, giving a lower continuum, but the chromospheric emission is unchanged. We note that there are two outliers, found in the two dark blue points on the left of the graph, likely indicating an erroneous B-V or T_{eff} value.

Using a similar approach, we attempt to get a value of S index on the Ca II IRT, only using the expression:

$$S \simeq \frac{F_L^m + F_R^m + F_T^m}{F_P^m + F_L^m}$$

Working again with mean fluxes, and assuming the α constant to be the same as for H & K. We use the same type of bandpasses of FWHM of 1.09 Å, situated on the lines at 8498.02 Å, 8542.09 Å, 8662.14 Å, and with continuum of 20 Å, around 8482 Å and 8710 Å. We also apply the same linear fit to that data, to obtain values in the same range we did previously. It is important to note that none of the spectra obtained show any sign of re-emission, even though the absorption is lesser when a higher re-emission is present in the H & K region, indicating that the IRT is also affected by chromospheric activity.

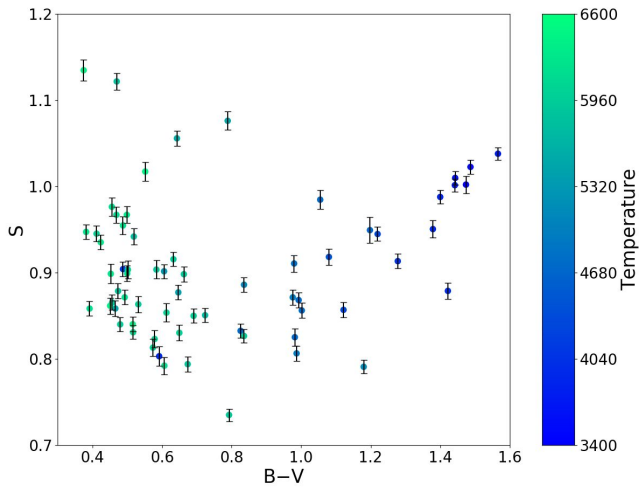


Fig. 2: *Measured S(I,R,T) scaled to the Mount-Wilson values as a function of B-V.*

On the S(I,R,T) graph (**Fig. 2**) the trend of S with the B-V (and by extension temperature) doesn't appear anymore, as the continuum flux is closer to be equal for all of the sample stars. This indicates that the IRT should be a better indicator of activity than the H & K, as it is much less dependant on the spectral class of the target star.

Furthermore, we propose here on **Fig. 3** the plot of S(H,K) against S(I,R,T), and we notice that:

- Because of the continuum variations on the bluer side, the S(H,K) give values with much more amplitude, whereas the S(I,R,T) has a shorter, more reasonable range.
- It appears that for hotter stars (with higher blue continuum) there is a relation between the S(H,K) and S(I,R,T), as only the cold stars produces high S(H,K) points.

III. b. Passage to $log(R'_{HK})$

The values obtained through this method still need some refining, as it still contains unwanted flux from the photosphere of the star itself. To get rid of this, we want to get the $log(R'_{HK})$, which is obtainable using the relation:

$$R'_{HK} = R_{HK} - R_{phot}$$

The R_{HK} is a first step of refining the S, getting rid of the previously seen effect, where S augments with colder stars. The R_{phot} is the the photospheric contribution to the flux measured in the bandpasses, which we want to get rid of as well.

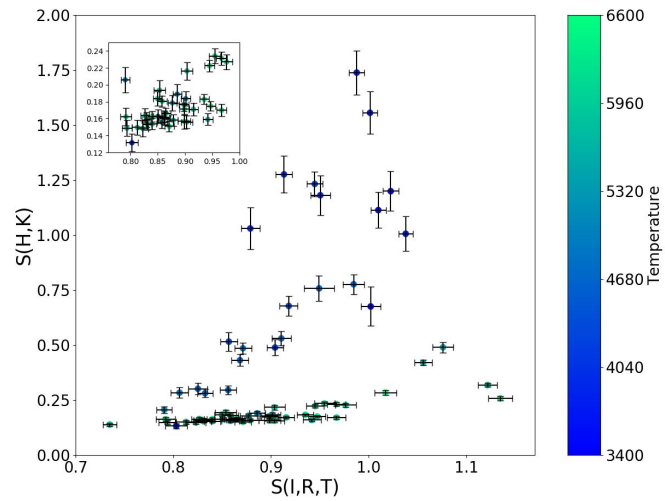


Fig. 3: *The obtained S(H,K) and S(I,R,T) graphed against each other. There seems to be a trend between the two for the hotter stars.*

Previous studies conveniently provide these values, based on functions of B-V. First we look at:

$$R_{HK} = 1.34 \cdot 10^{-4} \cdot C_{cf} \cdot S$$

Where the 1.34×10^{-4} factor and C_{cf} are derived in Middelkoop (1982). The values of the $log(C_{cf})$ are calculated beforehand using the bolometric luminosity and apparent magnitude of the stars in the solar neighborhood (Vaughan and Preston 1980).

$$log(C_{cf}) = 1.13(B-V)^3 - 3.91(B-V)^2 + 2.84(B-V) - 0.47$$

And for the R_{phot}, we have:

$$log(R_{phot}) = -4.898 + 1.918(B-V)^2 - 2.893(B-V)^3$$

Which was obtained by Hartmann et al. (1984). It is important to remark that this correction is going to have a much greater impact on chromospherically inactive stars, as their fluxes are going to be increased greatly. This relation is valid for the S(H,K) correction, but needs to be changed for the S(I,R,T), as the method used to determine C_{cf} uses the values of the continuum around the bandpasses.

Furthermore, we use the relation proposed by Sousa et al. (2008) to take into account the intrinsic metallicity of stars. We however propose a new fit linking the metallicity to the

T_{eff}, as our values are out of the range used in Sousa et al. (2008), we find:

$$T_{eff} = 7325 - 2456(B-V) + 368[\tfrac{Fe}{H}]$$

to be the best fit on our data, correlating much better than the original T_{eff} function (**Fig. 4**).

We note the fact that other possible transition functions were proposed, notably in Mittag 2013, using a C_{cf} and a T_{eff} expression from Rutten (1984) and Gray (2005). We then proceed to implement this into the metallicity correction, which gives us:

$$R'_{HK,new} = R'_{HK} \cdot (1 + \frac{368[Fe/H]}{7325-2456(B-V)})^4$$

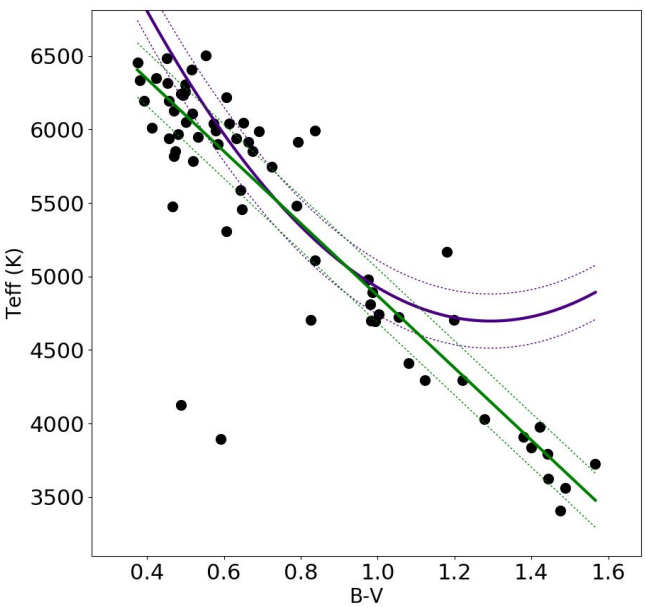


Fig. 4: *We find on this graph of T_{eff} against B-V, that the relation established in the Sousa et al. (2008) (purple) doesn't follow the trend on our data, which is why we propose this new fit (green).*

These corrections are usually enough to get clean and precise values, which is why we did not go further in the correction process. As previously explained, the corrections to get the R'_{HK} used previously cannot be directly applied to the S(I,R,T), as the C_{cf} function depends on the values measured in the continuum (Middelkoop 1982). This means we have to derive a new C_{cf} function to be able to extrapolate a $log(R'_{IRT})$.

III. c. The case of $log(R'_{IRT})$

The use of the Ca II IRT as an activity indicator has been attempted many times, and does show correlation with the H & K fluxes (Chmielewski 2000, Andretta et al. 2005, Busà et al. 2007, Martin et al. 2017, Martínez-Arnáiz et al. 2018). However, the methods used until now are model based, simulating the estimated absorption line, comparing the fluxes, and using the excess flux as re-emission. While this approach seems to show a strong link between the excess IRT flux to the H & K flux, the fact that it is model dependant could introduce errors, or even be biased. We will attempt to extract a value of the $log(R'_{IRT})$ based solely on the S(I,R,T) and a correction similar to that used in the $log(R'_{HK})$.

Do do this, we use the relation established by Middelkoop (1982), that allows us to calculate the C_{cf}, which we will then related to B-V:

$$C_{cf} \equiv (F_P + F_L) \cdot 10^{0.4\cdot(m_\gamma + BC)}$$

Where m_γ is the apparent magnitude of the star and BC the bolometric correction. We proceed in the same fashion: by getting the values of T_{eff} and BC from Johnson (1966) and establishing a new C_{cf} that will be used for the IRT region.

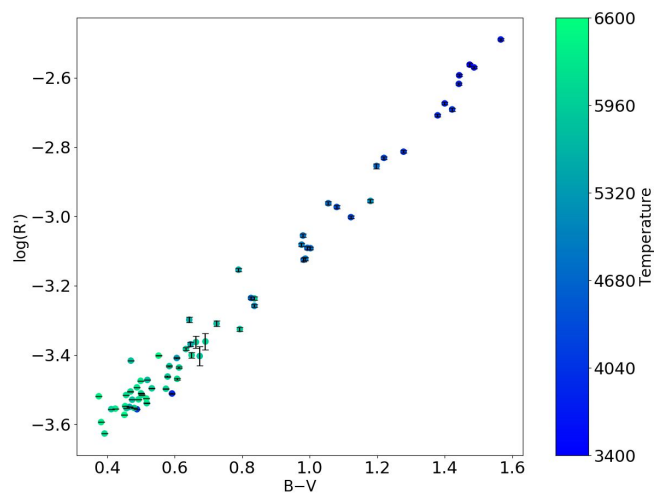


Fig. 5: *The values of the $log(R'_{IRT})$ compared to the B-V, we see that a strong relation exists with the temperature of the star.*

In doing this, we find that there exists a strong correlation between the B-V (and by extension the T_{eff}) of the star and the emission in the IRT region (**Fig. 5**). A simple linear fit on the data will be used to get rid of this unwanted trend, allowing us to do a visual check on the final product.

III. d. Photometry

The photometric data used comes from the Kepler C16 campaign, and are corrected using POLAR to take into account the errors from the spacecraft movement. We then proceed to calculate the RMS on the obtained light curves, giving us an idea of the variations of luminosity on those stars. More work is underway to provide an even better photometric variability value.

Now that all data has been processed, we can proceed with the analysis, where we look in detail at the $log(R'_{HK})$ and the $log(R'_{IRT})$, and compare the $log(R'_{HK})$ to the amplitude of photometric variability (hereafter reference to as pRMS).

IV. Results

IV. a. On the $log(R'_{HK})$

With all the corrections applied, we can now look at **Fig. 6**, which presents the $log(R'_{HK})$ against the B-V. We do find a plot that corresponds to the expected outcome, which shows a gap (the Vaughan-Preston gap) at -4.75. We recognize the two stellar populations identified as active (above -4.75) and inactive (below -4.75).

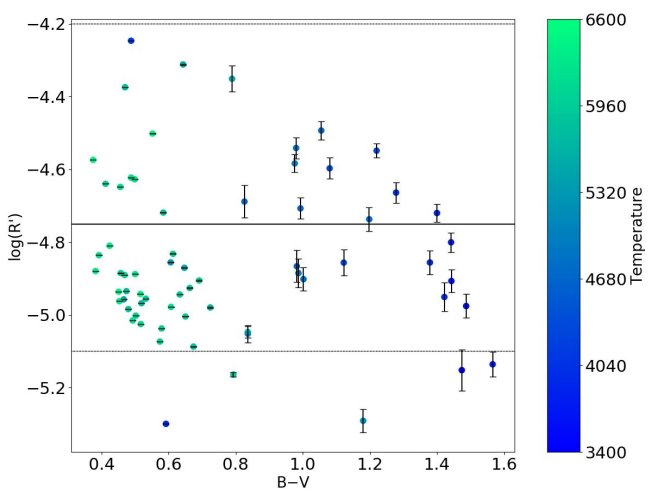


Fig. 6: *On this graph of the $log(R'_{HK})$ against the B-V, we can see the VP gap at -4.75. We also added the two other separators found in Henry et al. (1996), at -5.1 and -4.2 as the two dotted lines. These two lines were added to separate the "Very active" and "Very inactive" stars.*

IV. b. On the $log(R'_{IRT})$

Once the new corrections are applied to the $log(R'_{IRT})$, we obtain **Fig. 7**.

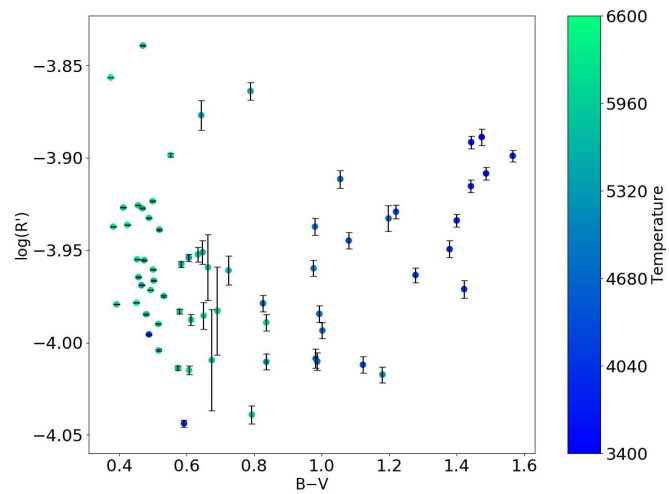


Fig. 7: *Values of the $log(R'_{IRT})$ against the B-V.*

Even though we do obtain a similar repartition to that of the $log(R'_{HK})$ in **Fig. 6**, the placements of the points in **Fig. 7** are not exactly the same, which can have multiple causes.

1. The correction functions used here (whether the new C_{cf} or the delinearization of the data) are wrong, and need better replacement.
2. The IRT fluxes are correlated to the H & K, but are not strong enough to be picked up by a similar approach as for the H & K.
3. The IRT fluxes are not well enough correlated with the H & K to obtain precisely matching graphs.
4. The IRT fluxes and H & K fluxes are actually not correlated.

Given the precision of the data and methods used to determine the new C_{cf}, option 1 is favored over the others. However, given the close results obtain through this mean, it is likely that with more accurate data, a viable correction can be found. In the event that option 2 is true, a model could be a viable way of extracting a $log(R'_{IRT})$, because the values seem close to being correct.

We find option 4 unlikely, as, even though model based, previous work have shown that a correlation does seem to take place between the two domain. The second most viable explanation is 3, meaning that the two regions are not correlated well enough to each other to be compared directly to one another. This idea seems to be reinforced by Martínez-Arnáiz et al. (2018), that shows the link between

the K line and the IRT lines; a linear relation can be seen between the two, however, it isn't a perfect fit.

IV. c. Comparing the activity of the $log(R'_{HK})$ with the photometric variability (pRMS)

Because the $log(R'_{HK})$ is a much more established and accepted activity indicator, we simply propose to compare it to the pRMS, shown in **Fig. 8**.

If we decompose the graph in 4 regions, namely:

I. High pRMS, low $log(R'_{HK})$
II. High pRMS, high $log(R'_{HK})$
III. Low pRMS, low $log(R'_{HK})$
IV. Low pRMS, high $log(R'_{HK})$

we observe that most of the stars are in regions II and III, which suggests a link between $log(R'_{HK})$ and the pRMS, as the pRMS grows with the $log(R'_{HK})$. Regions I and IV present stars that do not follow that rule. The total number of stars in those regions combined represent about 25% of the sample. Because this linear trend is not clear, it could turn out to be even higher.

Fig. 8: *Comparing the photometric variability amplitude with its proxy, the $log(R'_{HK})$. The VP gap is visible as the continuous line, the dashed line is the potential pRMS gap observed at 0.4 millimag.*

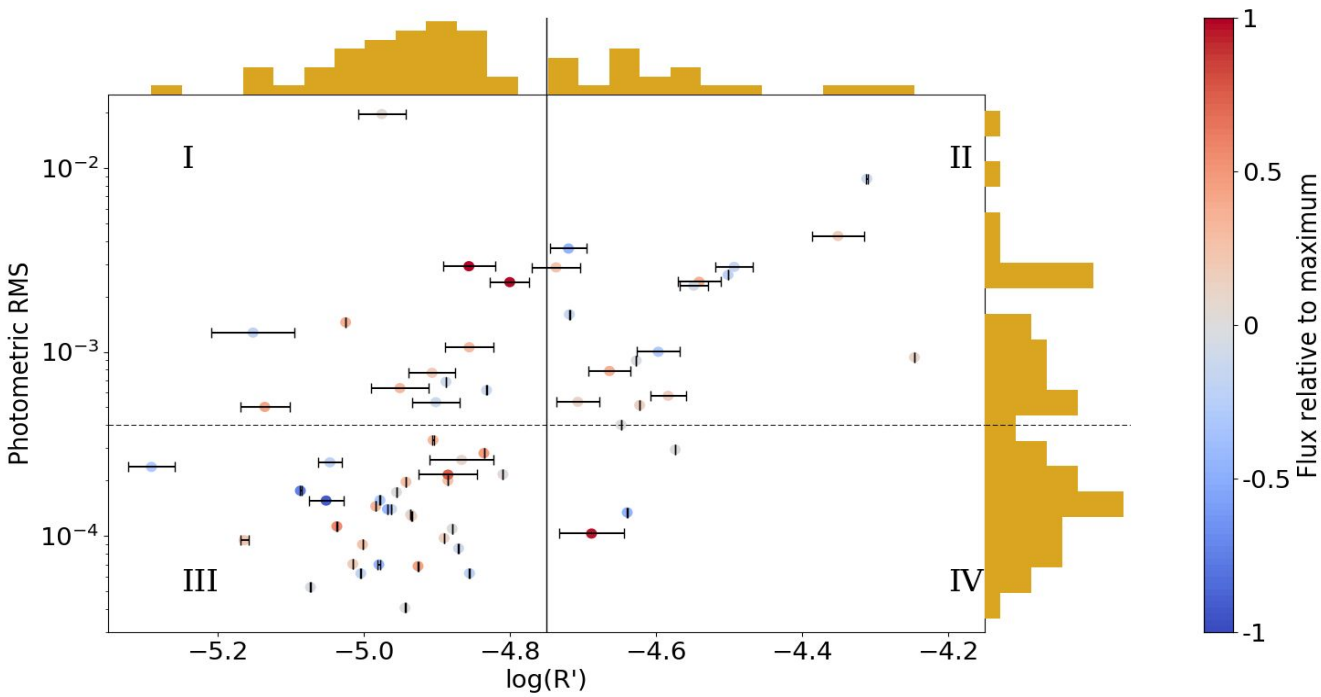
This indicates that under -4.6, the $log(R'_{HK})$ is not a good stellar activity indicator, as we can clearly see the stellar population diverging away from the original trend, shown in Lockwood et al. (2007). This trend was established using ground-based photometry, limiting detection in the low amplitude region. We note that in our data, a correlation between the $log(R'_{HK})$ and the pRMS seems to appear for $log(R'_{HK}) \succ -4.6$.

In addition, the color scale indicates for each star its current activity level in reference to its own variability, established with its pRMS. As is evident in the graph, this activity level tells us that intrinsic activity is not what forms the gap. If that was the case, we would only find stars at the top of their activity above the VP gap, and stars at their lowest activity under.

V. Discussion

Finally, we remark the presence of a potential gap in the pRMS, similar to the VP gap for the $log(R'_{HK})$. It is unclear if this gap is real or simply a byproduct of our own data, but in both cases a more detailed study should be made to verify its legitimity.

From **Fig. 8**, we can extrapolate different regimes of stellar activity, determined by the type of features on the surfaces of these stars. Again using the four regions precedently defined, we can hypothesis those regimes.



- Region I corresponds to a population of spots dominated stars, inducing large pRMS, but not exciting the chromosphere as much.
- Region IV corresponds to plages dominated stars, not varying much in pRMS, but heating the chromosphere more.
- Region II and III are a mix of the two, with region II having a lot of both, and region III few of both.

These assumptions are based upon the fact that spots induce much more pRMS variations that plages, demonstrated in articles such as Shapiro et al. (2014), Dumusque et al. (2014), Shapiro et al. (2016) or Shapiro et al. (2017).

In any case, this study reveals that the $log(R'_{HK})$ is not as good an indicator as we previously believed. It is for that reason that we recommend extreme precautions when selecting low activity targets. The active part of the population shouldn't be too affected by this, as the trend between the $log(R'_{HK})$ and the activity level of the stars still seem linked. However, when it comes to the population of less active stars, it is not safe to assume that a low $log(R'_{HK})$ ensures a low activity level.

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